

Study details

Authors

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Abstract

Alcohol consumption leads to formation of phosphatidylethanol (PEth) via the transphosphatidyl-lation activity of phospholipase D (PLD) enzymes. Though this non-natural phospholipid routinely serves as a biomarker of chronic alcoholism, its pathophysiological roles remain unknown. We use a minimalist diazirine alkyne alcohol as an ethanol surrogate to generate clickable, photoaffinity lipid reporters of PEth localization and lipid–protein interactions via PLD-mediated transphosphatidylation. We use these tools to visualize phosphatidyl alcohols in a manner compatible with standard permeabilization and immunofluorescence methods. We also use click chemistry tagging, enrichment, and proteomics analysis to define the phosphatidyl alcohol interactome. Our analysis reveals an enrichment of putative interactors at various membrane locations, and we validate one such interaction with the single-pass transmembrane protein basigin/CD147. This study provides a comprehensive view of the molecular interactions of phosphatidyl alcohols with the cellular proteome and points to future work to connect such interactions to potential pathophysiological roles of PEth.

Lipid probes utilized

Phosphatidyl Alcohol

Cell line analyzed

[HeLa](#)

Species: *Homo sapiens*

Cell type: Epithelial (immortalized cervical cancer line)

Uncaging & Crosslinking timeline

Mass spectrometry quantification method

PSM

Data wrangling

Use the download button below to download the R script used to wrangle the authors' original submission into the data visualization tools on the Lipid Interactome:

Use the download button below to download the original data file prior to wrangling: